Astro HW 11

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11.1

a) Spiral galaxies - Tracing the Light

$$L(r) = \int_0^r 2\pi s \Sigma(s) ds = 2\pi h_r^2 \left[1 - \left(\frac{r}{1+hr}\right) e^{-r/h_r}\right]$$

$$L(r) = \int_0^R 2\pi r \left[\Sigma_0 exp(-r/h_r)\right] dr$$

$$L(r) = 2\pi\Sigma_0 \int_0^R r \left[exp(-r/h_r)\right] dr$$

$$x = r/h_r, \quad xh_r = r, \quad dxh_r = dr$$

$$= 2\pi\Sigma_0 h_r^2 \int_0^R xe^{-x} dr$$

$$= 2\pi\Sigma_0 h_r^2 \left[{}_0^R - (r/h_r + 1)e^{-r/h_r} + C\right]$$

$$L(r) = 2\pi\Sigma_0 h_r^2 \left[1 - (R/h_r + 1)\right] e^{-R/h_r}$$

b)

Given ...

$$\Sigma = 36 L_\odot \text{pc}^{-2}, \qquad h_r = 4\text{kpc}, \qquad R_0 = 8.5\text{kpc}$$

$$\Sigma = \Sigma_0 \exp(-r/h_r)$$

We find ...

$$\Sigma_0 = 301.42 L_\odot \text{pc}^{-2}$$

c)

$$L(r) = 2\pi\Sigma_0 h_r^2 1 - \left[\frac{(R/h_r + 1)}{e^{R/h_r}}\right]$$

$$L(R) = 2\pi\Sigma_0 h_r^2 \lim_{R\to\infty}\left[\frac{(R/h_r)}{e^{R/h_r}}\right]$$

Apply L'hoptials ...

$$L(\infty) = 2\pi\Sigma_0 h_r^2 \lim_{R\to\infty}\left[\frac{(1/h_r)}{Re^{R/h_r}}\right]$$

$$L(\infty) = 2\pi\Sigma_0 h_r^2$$

$$\boxed{L(\infty) \approx 3.030 \times 10^{10}\,\mathrm{L}_\odot}$$

d)

Given...

$$L/3 = M_{mw}, \qquad L = 3.030 \times 10^{10}\,\mathrm{L}_\odot$$

We find ...

$$\boxed{M_{mw} \approx 1.010 \times 10^{10}\,\mathrm{M}_\odot}$$

e)

Given ...

$$L(R_0)/L(\infty), \qquad L(\infty) = 3.030 \times 10^{10}\,\mathrm{L}_\odot$$
$$L(r) = 2\pi\Sigma_0 h_r^2 \left[1 - (R/h_r + 1)\right] e^{-R/h_r}$$

We find ...

$$L(R_0) = 1.899 \times 10^{10}\,\mathrm{L}_\odot\mathrm{pc}^{-2}$$
$$L(R_0)/L(\infty) = 62\%$$

11.2 Spiral galaxies - Tracing the Matter

a)

Given...

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

$$v_{esc} = 500\mathrm{kms}^{-1} \qquad 8.5\mathrm{kpc}$$

We find that...

$$v_{esc} = \frac{\sqrt{2GM}}{\sqrt{R}}$$

$$\frac{v_{esc}\sqrt{R}}{\sqrt{2G}} = \sqrt{M}$$

$$M = 2.470 \times 10^{11}\,\mathrm{M}_\odot$$

b)

i)

Given ...

$$M_V = -2.76 \log_{10}\left(\frac{P}{10\mathrm{d}}\right) - 4.16$$

$$m_V = 18.9, \qquad P = 31.47\mathrm{d}$$

$$m - M = 5\log_{10}\left(\frac{d}{10\,\mathrm{pc}}\right) \to d = 10\exp\left(\frac{m-M+5}{5}\right)\ \mathrm{pc}$$

We find ...

$$\boxed{M_v = -5.534, \qquad d = 7.706\times10^{-1}\,\mathrm{Mpc}}$$

ii)

Given ...

$$v = H_0 d$$

$$d = d = 7.706\times10^{-1}\,\mathrm{Mpc}$$

We find ...

$$\boxed{53.9\mathrm{km/s}}$$

iii)

Given ...

$$\frac{\Delta v}{\Delta T} = a$$

$$v_{init} = 53.9\mathrm{km/s}, \qquad v_{final} = -119\mathrm{km/s}, \qquad \Delta v = v_{final} - v_{init} \qquad \Delta T = 14\mathrm{Gyr}$$

$$\boxed{g = -3.914\times10^{-13}\,\mathrm{m/s}}$$

iv)

Given ...

$$a = GM/R^2$$

$$g_{M31} = GM_{m31}/R^2, \qquad g_{MW} = GM_{Mw}/R^2$$

$$R = 7.706\times10^{-1}\mathrm{Mpc}$$

$$a = a_m + a_{m31}$$

We can find ...

$$\cancel{-}3.914\times10^{-13} = \cancel{-}3GM_{MW}/R^2$$

$$\frac{(3.914\times10^{-13})R^2}{3G} = M_{MW}$$

$$\boxed{M_{MW} = 5.559\times10^{11}\,\mathrm{M}_\odot}$$

11.3

a)

Given ...

$$R = D\theta$$

$$D = 16.8\,\mathrm{Mpc}, \qquad \theta = .1''$$

We find...

$$\boxed{R = 8.144\mathrm{pc}}$$

b)

From figure ...

$$-.1'' = 1780\,\mathrm{km/s}$$

$$+.1'' = 650\,\mathrm{km/s}$$

We find ...

$$\boxed{c_{vel} = 1215\,\mathrm{km/s}}$$

c)

Given ...
Assuming circular orbit ...

$$v^2 = \frac{GM}{R}$$

$$M = \frac{v^2 R}{G}$$

$$R = 8.144\mathrm{pc}$$

We find ...

$$\boxed{M_{bh} = 4.613 \times 10^{38}\,\mathrm{kg} = 2.320 \times 10^{8}\,\mathrm{M}_{\odot}}$$

11.4 Redshifts and the Hubble Law

a)

Given ...

$$Z = \frac{\lambda_{obsv} - \lambda_{rest}}{\lambda_{rest}}$$

$$\lambda_{rest} = 486.135\,\mathrm{nm}, \qquad \lambda_{obsv} = 497.97\,\mathrm{nm}$$

We find ...

$$\boxed{Z = .023}$$

b)

Given ...

$$m - M = 5\log(d/10\text{pc})$$
$$v = H_0 d, \qquad z \simeq \frac{v}{c}, \qquad \rightarrow \frac{cz}{H_0} = d$$
$$H_0 = 70\,\text{km/s/Mpc}$$

With the goal of deriving ...

$$m - M = 5\log(z) + 43.2$$

We can do ...

$$m - M = 5\log_{10}\left(\frac{d}{10\,\text{pc}}\right)$$
$$m - M = 5\log_{10}\left(\frac{\frac{cz}{H_0}}{10\,\text{pc}}\right)$$
$$m - M = 5\log_{10}\left(\frac{\frac{cz}{H_0}}{10^{-5}\,\text{Mpc}}\right), \qquad m - M = 5\log_{10}\left(\frac{cz}{H_0}\frac{10^5}{\text{Mpc}}\right)$$
$$m - M = 5\log_{10}\left(\frac{c}{H_0}\frac{10^5}{\text{Mpc}}\right) + 5\log_{10}(z)$$
$$5\log_{10}\left(\frac{c}{H_0}\frac{10^5}{\text{Mpc}}\right) \approx 43.15$$

Hence ...

$$\boxed{m - M = 5\log_{10}(z) + 43.2}$$

c)

Given ...

$$m - M = 5\log_{10}(z) + 43.2$$
$$Z = .023, \qquad mV = 12.7$$

We find ...

$$\boxed{M_v = -22.379}$$

d)

It is more likely to be an elliptical galaxy, as they have a absolute magnitude of ≈ -22

11.5 Extra Credit

a)

It is a spiral, it has new, 'infantile', stars, with the also nebulae becoming structured. It also is spinning!

b)

Given ...

$$V = H_0 * D$$

$$D = 2,000,000\text{mi}$$

We find ...

$$V = 2.684 \times 10^{-13}\,\text{km/s}$$

This means that it is barely moving away from us, whereas the poem says that it is receding 'At several million miles per sec', thus that the author is taking artistic liberty.

c)

It is probably before 1929, because Hubble discovered the proper recession then.